Herman and Alpher calculated and proposed a temperature of 5 degrees Kelvin for the universe. Well, that's just plain wrong. The precisely measured temperature of these microwaves is 2.725 degrees, sometimes written as simply 2.7 degrees, and if you're numerically lazy, nobody will fault you for rounding the temperature of the universe to 3 degrees.

Let's pause for a moment. Herman and Alpher used atomic physics freshly gleaned in a lab, and applied it to hypothesized conditions in the early universe. From this, they extrapolated billions of years forward, calculating what temperature the universe should be today. That their prediction even remotely approximated the right answer is a stunning triumph of human insight. They could have been off by a factor or ten, or a hundred, or they could have predicted something that isn't even there. Commenting on this feat, the American astrophysicist J. Richard Gott noted, "Predicting that the background existed and then getting its temperature correct to within a factor of 2, was like predicting that a flying saucer 50 feet wide would land on the White House lawn, but instead, a flying saucer 27 feet wide actually showed up."

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The first direct observation of the cosmic microwave background was made inadvertently in 1964 by American physicists Arno Penzias and Robert Wilson of Bell Telephone Laboratories, the research branch of AT&T. In the 1960s everyone knew about microwaves, but almost no one had the technology to detect them. Bell Labs, a pioneer in the communications industry, developed a beefy, horn-shaped antenna for just that purpose.

But first, if you're going to send or receive a signal, you don't want too many sources contaminating it. Penzias and Wilson sought to measure background microwave interference to their receiver, to enable clean, noise-free communication within this band of the spectrum. They were not cosmologists. They were techno-wizards honing a microwave receiver, and unaware of the Gamow, Herman, and Alpher predictions.

What Penzias and Wilson were decidedly not looking for was the cosmic microwave background; they were just trying to open a new channel of communication for AT&T.

Penzias and Wilson ran their experiment, and subtracted from their data all the known terrestrial and cosmic sources of interference they could identify, but one part of the signal didn't go away, and they just